1 Why Does MSE Underperform When One Action Suffices?

Generative robot policies are often motivated by their ability to represent multiple valid actions for the same observation. If those actions follow incompatible strategies, a policy trained with mean squared error (MSE) may average them into an invalid motion. This explanation predicts that repeated samples from a successful policy should reveal distinct action strategies. We test this prediction by holding the observation and instruction fixed and sampling the policy repeatedly.

The continuous actions favor a single Gaussian description. We probe GR00T on GR1 and $\pi_{0.5}$ on LIBERO using their evaluation-time samplers. For each policy, we screen 3,000 states and retain those with the most separated samples. We then draw 32 fresh action chunks per retained state and compare a single Gaussian with a two-component Gaussian mixture along an independently estimated principal direction. The analysis covers GR1's arm and waist joints and LIBERO's six arm channels. Table 1 shows that the median held-out likelihood favors the single Gaussian in both settings. Even this search for separated samples supports a simple description of continuous motion: variation around one local action.

Table 1: **Continuous actions favor a single Gaussian.** The 1G advantage is the median held-out log-likelihood difference between a single Gaussian and a two-component mixture, in nats per projected sample. Positive values favor the single Gaussian. States are selected from a 3,000-state screen to maximize apparent separation; each has 32 fresh samples.

Policy / benchmark	Channels	States	1G advantage
GR00T / GR1	Arm + waist	30	$+0.331$
$\pi_{0.5}$ / LIBERO	Arm	36	$+0.352$

Averaging the sampled actions also preserves the local motion. At 40 WidowX states, we estimate the arm-action mean from 64 samples, restore the simulator state, and execute that mean over the four control steps used before replanning. Each comparison holds the gripper commands fixed. The resulting motion stays inside the empirical 95% envelope of sampled paths at all 40 states. Thus, in this direct intervention, averaging produces an ordinary local motion. A regression policy has a useful action to predict.

Yet MSE regression can still perform substantially worse than Flow. Table 12 of Pan et al. (2025) reports a clear gap on state-based manipulation tasks with the same backbone and training data. With Sudeep-DiT, best-checkpoint success is 12% for regression versus 40% for Flow on Transport-mh, and 52% versus 86% on Tool-Hang, averaged over three seeds. The Transport-mh gap also appears with Chi-Transformer and Chi-UNet. The difficulty therefore persists even when regression uses the same architecture as the generative policy.

These observations pose a simple question: *if one local action suffices, why is MSE an ineffective way to learn it?* MSE targets the conditional mean, making it a natural objective for a Gaussian action model. The unresolved issue is how the training loss fits that prediction across the dataset. We next examine two properties that shape this process: action distributions with different radii at different task stages, and large errors from rare or noisy actions. These properties motivate a loss that adapts to both local scale and heavy tails.

REFERENCES

Chaoyi Pan, Giri Anantharaman, Nai-Chieh Huang, Claire Jin, Daniel Pfrommer, Chenyang Yuan, Frank Permenter, Guannan Qu, Nicholas Boffi, Guanya Shi, and Max Simchowitz. Much ado about noising: Dispelling the myths of generative robotic control. *arXiv preprint arXiv:2512.01809*, 2025. URL `https://arxiv.org/abs/2512.01809`.